



Inspiring Excellence

CSE 251

Electronic Devices and

Circuits

Lecture 2

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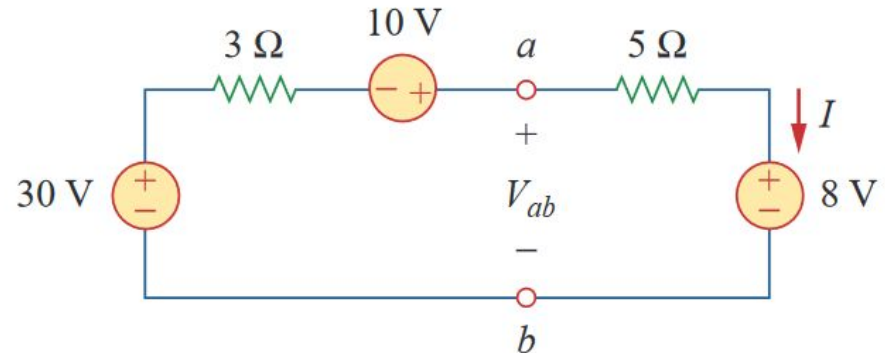
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Alternative Circuit Representation: Line diagrams

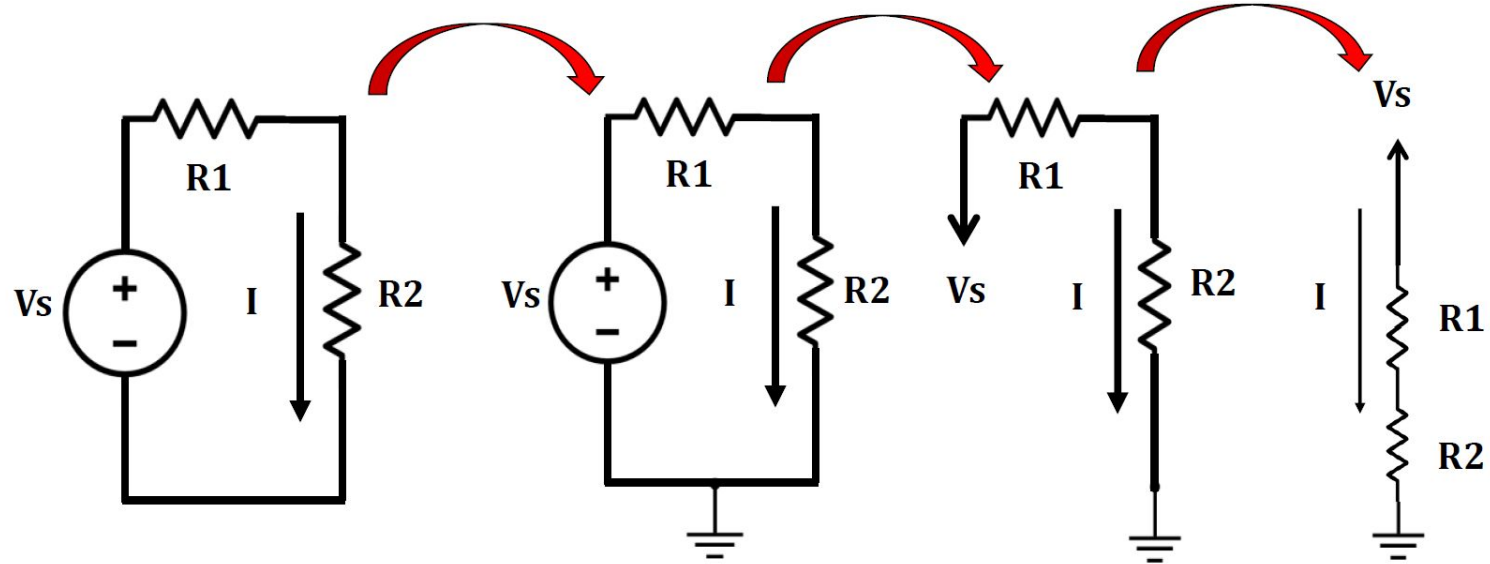
Steps to decompose circuits to line diagram

1. Set a ground so that number of **floating voltage** sources are minimized. **Floating voltage** sources:
2. Detach the ground **from the voltage source**.
3. Convert the non-floating voltage sources (~~current sources~~) into:
 - Arrow : ($\rightarrow$) **Fixed/Constant voltage source**
4. Keep passive elements as they are.

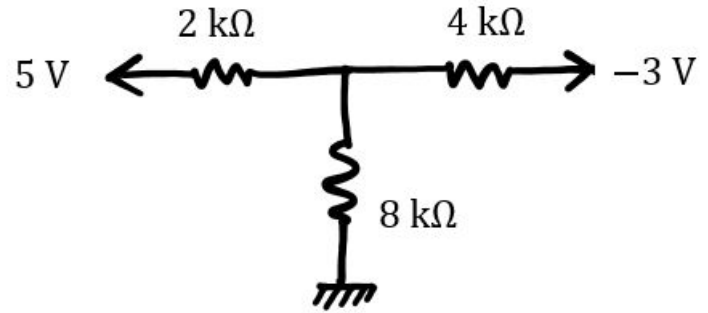
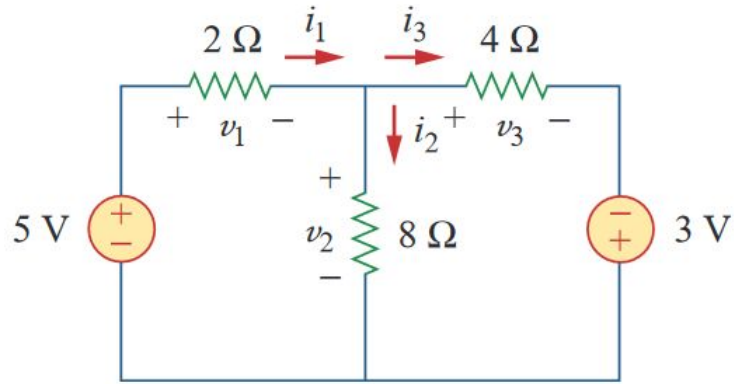
Voltage sources which are **not connected the ground** terminal. In the diagram, the **10 V** voltage source is floating



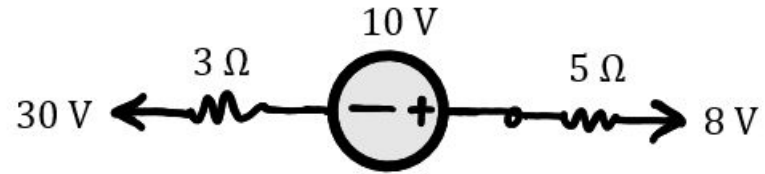
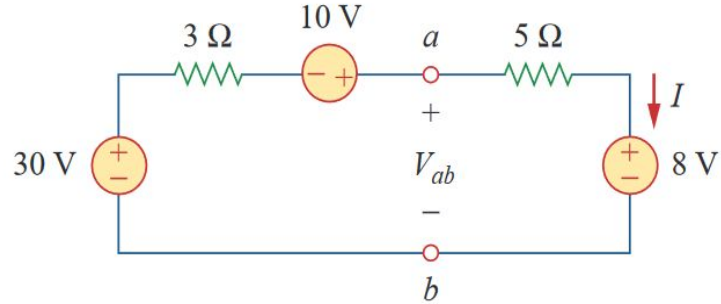
Line Diagram: Example 1



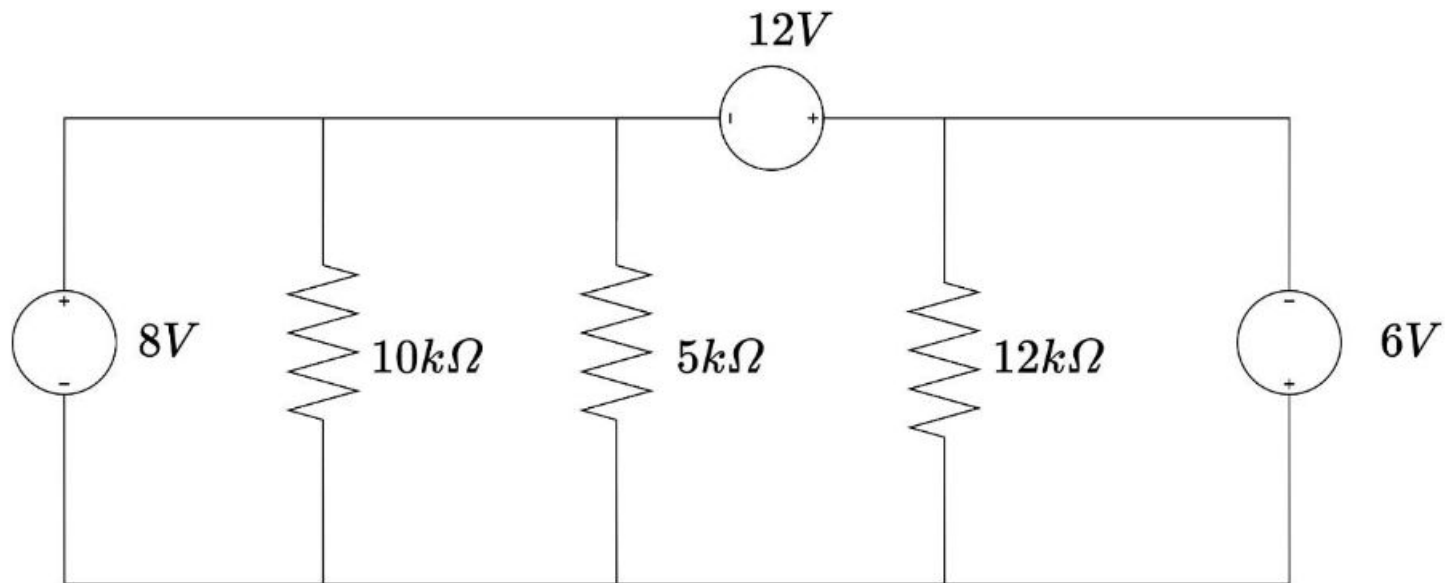
Line Diagram: Example 2



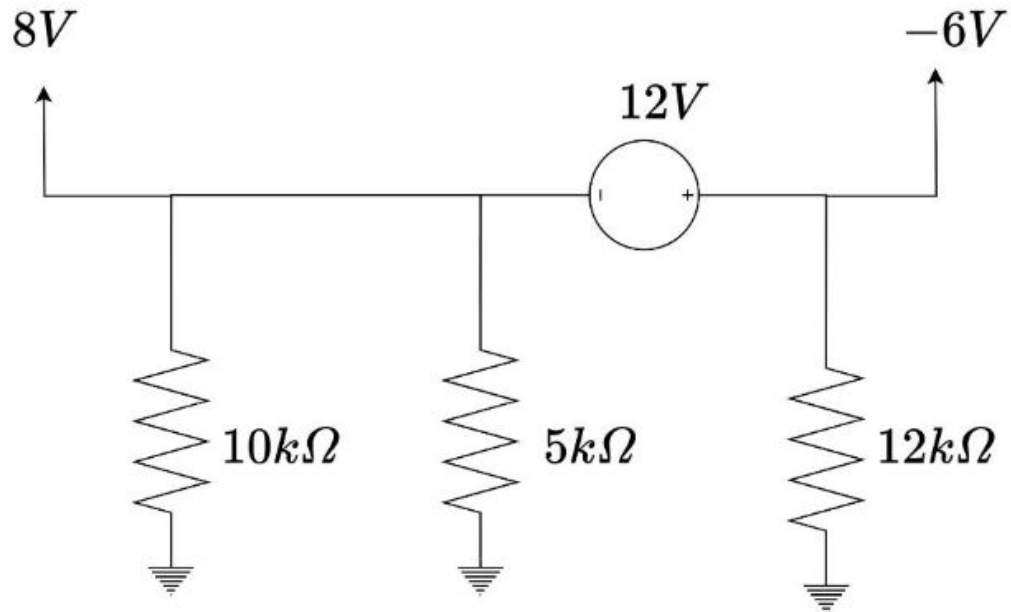
Line Diagram: Example 3



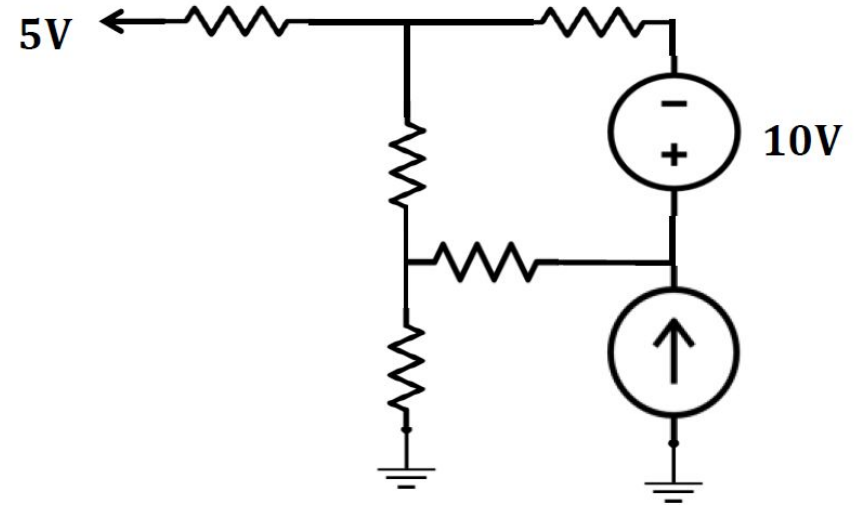
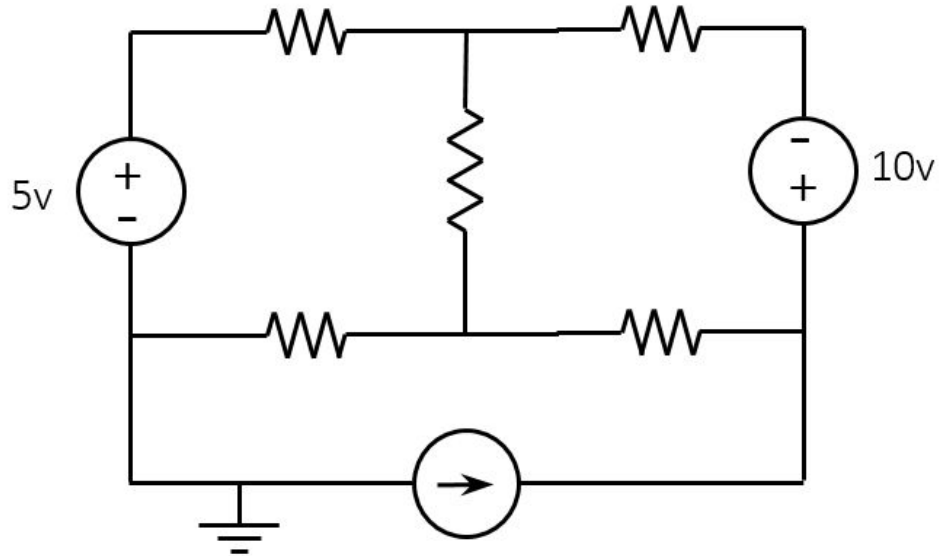
Line Diagram: Example 4



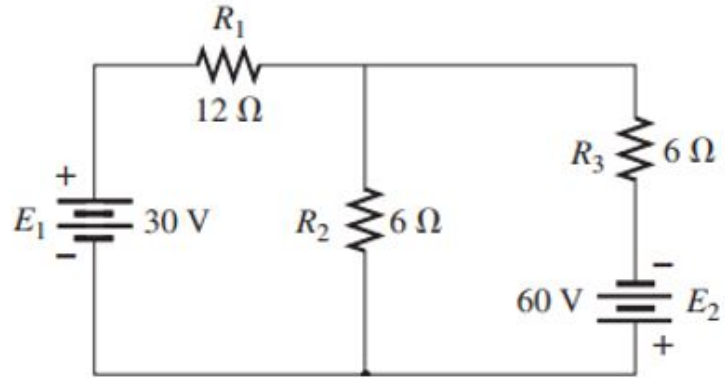
Line Diagram: Example 4



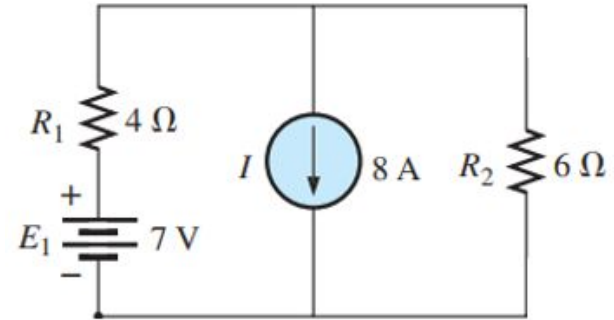
Line Diagram: Example 5



More Examples

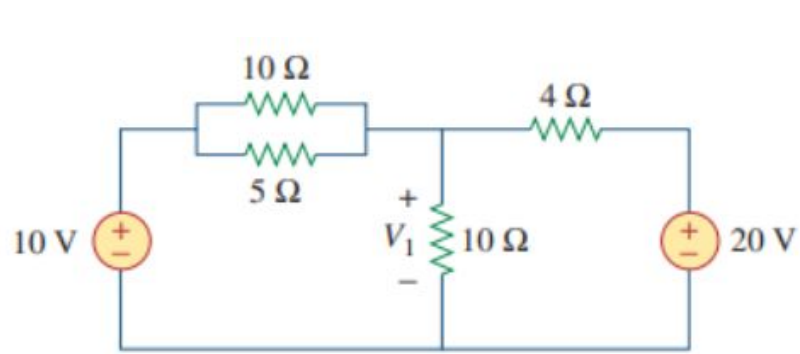


Circuit 1

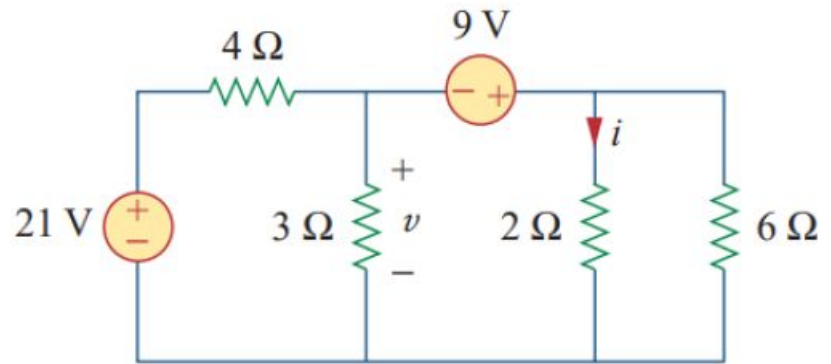


Circuit 2

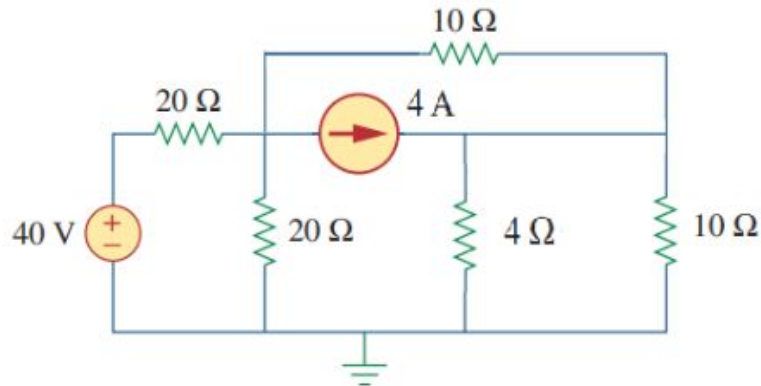
Practice Problems



Circuit 1



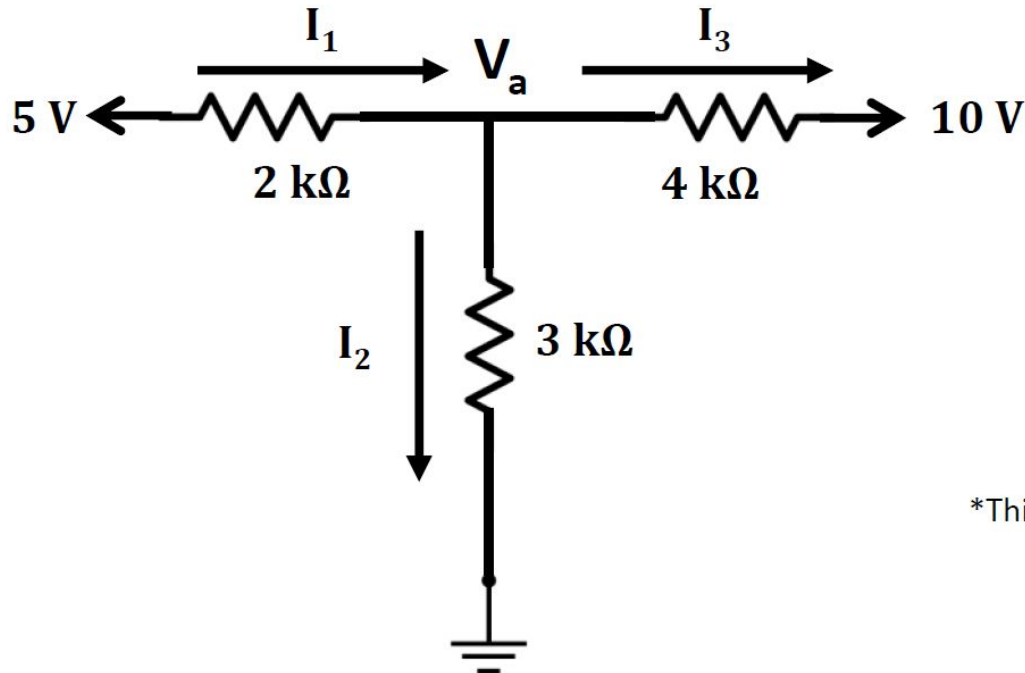
Circuit 2



Circuit 3

Kirchhoff's Current Law (KCL):

- "The algebraic sum of all currents entering and exiting a node must equal zero."
- "Currents flowing into a node (or a junction) must be equal to the currents flowing out of it."



$$I_1 = I_2 + I_3$$

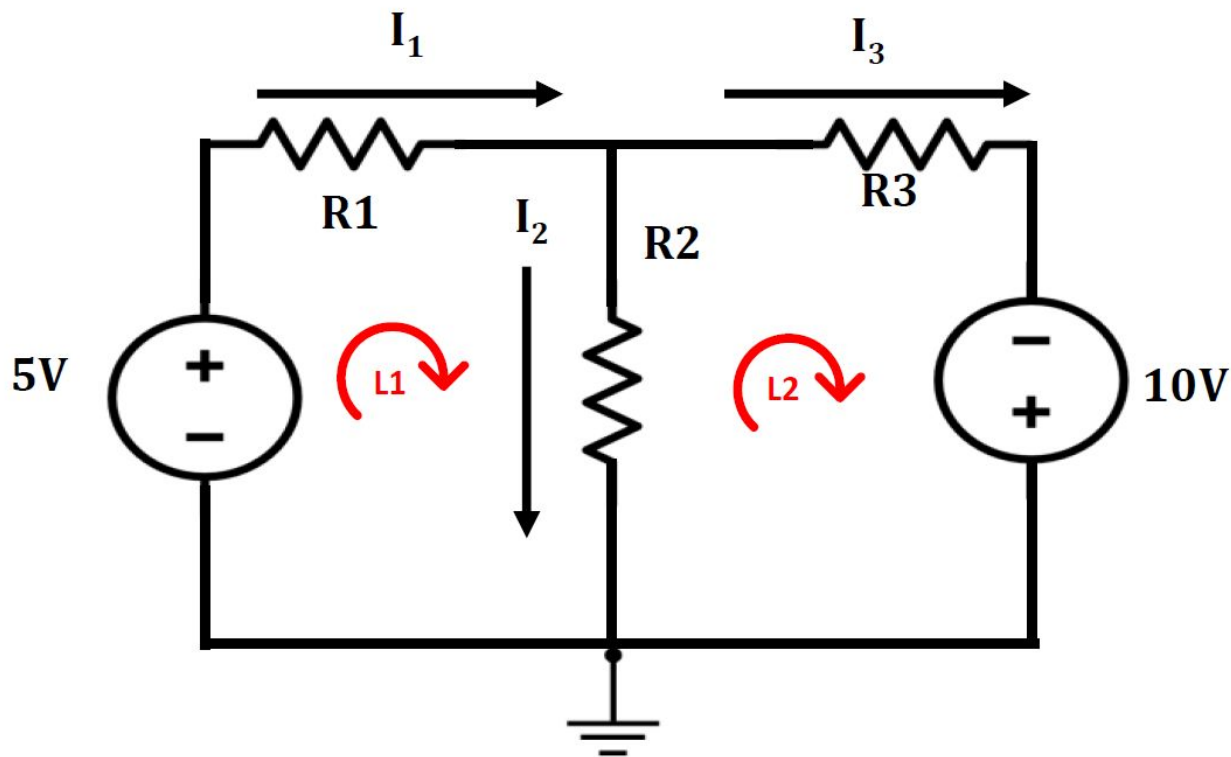
*This is also applicable for supernodes!

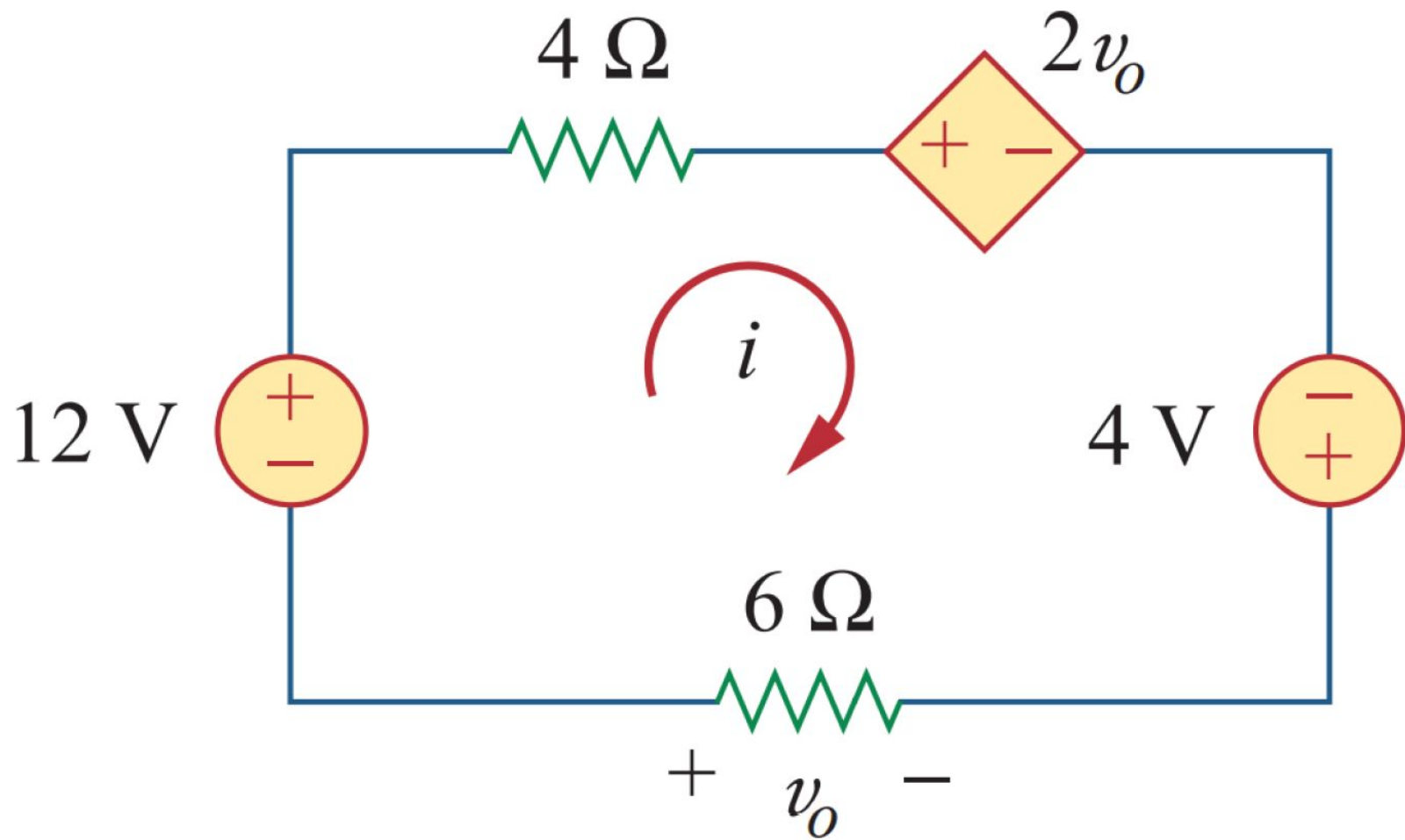
Kirchhoff's Voltage Law (KVL): The algebraic sum of all voltages in a loop must equal zero

$$\text{Loop 1: } 5 - I_1 R_1 - I_2 R_2 = 0$$

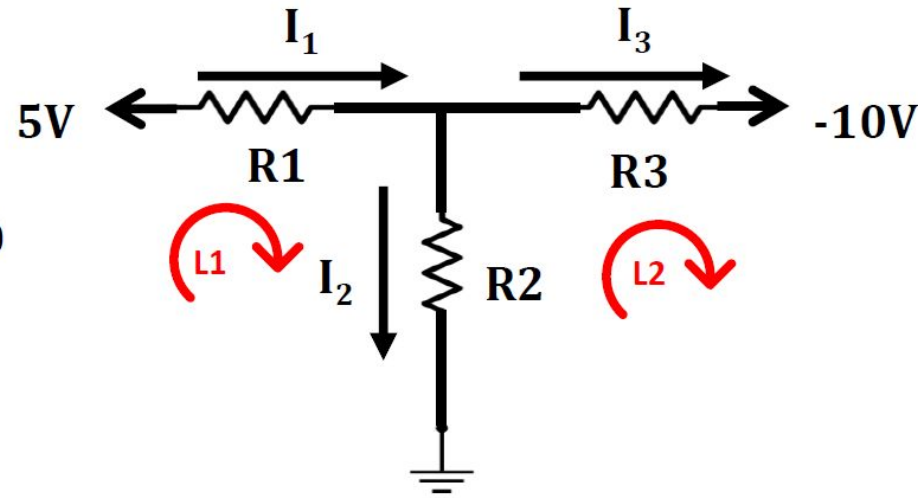
$$\text{Loop 2: } 10 + I_2 R_2 - I_3 R_3 = 0$$

$$\sum V = 0$$





$$\sum V [\text{along line}] = \text{Voltage at the starting of the node} - \text{Voltage at the ending of the node}$$

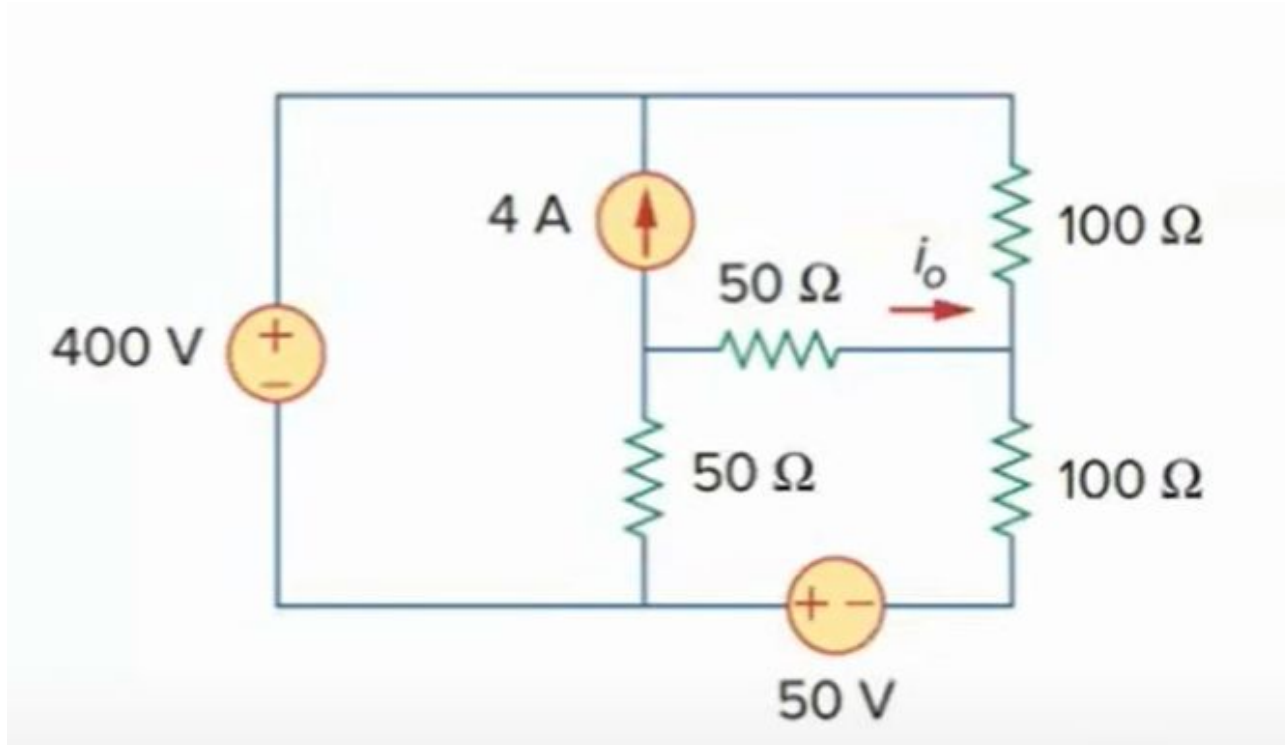


Line 1: $5 - 0 = I_1 R_1 + I_2 R_2 = 0$

Line 2: $0 + 10 = -I_2 R_2 + I_3 R_3$

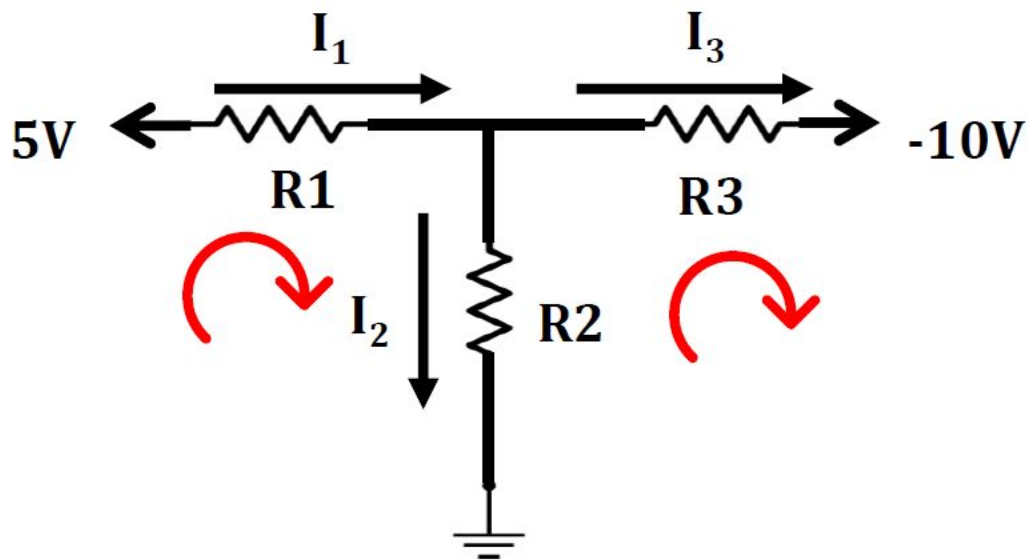
Practice Problem

Draw the alternative representation of the following circuit minimizing the number of floating voltage sources
Find i_o using KVL



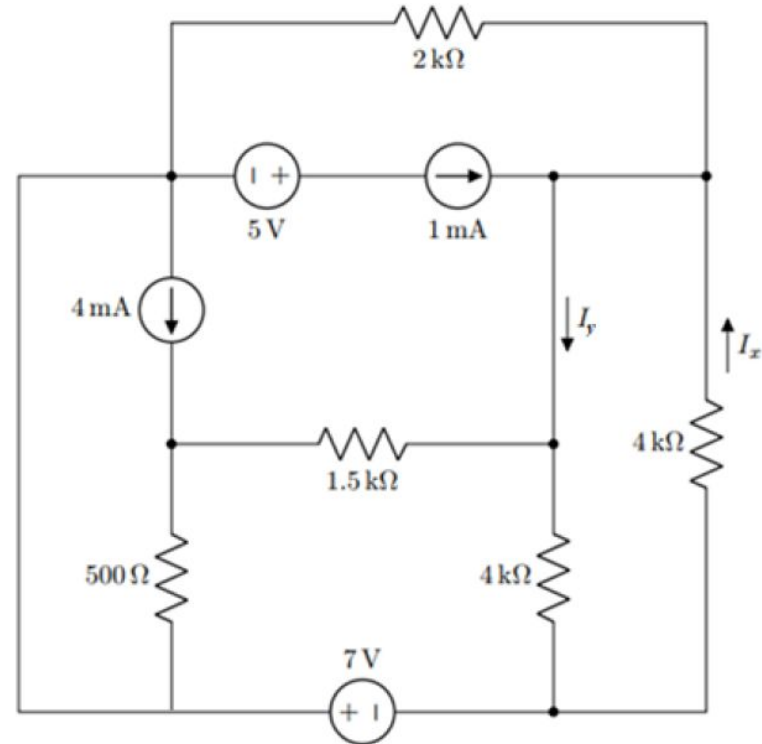
Nodal analysis:

$$V_a \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) - \frac{5}{R_1} - \frac{0}{R_2} - \frac{-10}{R_3} = 0$$



Practice Problem

Draw the alternative representation of the following circuit minimizing the number of floating voltage sources. Also find i_x and i_y



Practice Problem

Draw the alternative representation of the following circuit minimizing the number of floating voltage sources.

Find i_x using KVL

